The documentation and process conversion measures necessary to comply with this revision shall be completed by 11 February 1998

INCH-POUND

MIL-PRF-19500/619A
11 November 1997
SUPERSEDING
MIL-S-19500/619
11 May 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, SCHOTTKY RECTIFIERS,
JAN, JANTX AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for schottky rectifiers intended for use in power surface mount applications. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (surface mount).

1.3 Maximum ratings.

Column 1	Column 2	Column 3	Column 4	Column 5		Column 6	
Types 1/	Peak inverse voltage	I_O 2/ T_C = +100°C	I_{FSM} 3/	Maximum 4/ thermal resistance		T_{STG}	T_{OP}
	V dc	A dc	A(pk)	°C/W 5/	°C/W 6/	°C	°C
M19500/619-13Z	30	110	1,000	0.20	0.45	-65 to +175	-65 to +150
M19500/619-23Z	45	150	1,000	0.20	0.45		-65 to +175
M19500/619-33Z	100	120	900	0.20	0.45		-65 to +175
M19500/619-12Z	30	60	900	0.35	0.80		-65 to +150
M19500/619-22Z	45	75	900	0.35	0.80		-65 to +175
M19500/619-32Z	100	60	900	0.35	0.80		-65 to +175
M19500/619-21Z	45	10	300	0.85	1.40		-65 to +175
M19500/619-31Z	100	10	300	0.85	1.40		-65 to +175

1/ See 1.5 for type designation/lead configuration.
2/ Derate the I_O linearly to zero at 100°C to the maximum operating temperature.
3/ I_{FSM}, t_p = 8.3 ms, T_C = +25°C, V_R = PIV.
4/ Maximum thermal resistance junction to cathode surface.
5/ Thermal resistance value for package outline (YZ) (i.e., 11, 21, 31, 12, 22, 32, 13, 23, and 33).
6/ Thermal resistance value for package outline (YZ) (i.e., 14, 24, 34).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics at T_C = 25°C.

Column 7	Column 8	Column 9	Column 10	Column 11	Column 12	Column 13	Column 14
Types	Maximum forward voltage per leg T_C = +25°C		Maximum forward voltage per leg T_C = +125°C		Maximum reverse current per leg at PIV mA		Maximum junction capacitance per leg 1 KHz ≤ f ≤ 1 MHz V_R = 5 V
	Volts	Amperes	Volts	Amperes	T_J = +25°C	T_J = +125°C	pF
M19500/619-13Z	0.53	110	0.43	110	7.5	400 1/	7,500
M19500/619-23Z	0.75	150	0.65	150	5.0	150	5,000
M19500/619-33Z	0.90	120	0.73	120	5.0	100	2,700
M19500/619-12Z	0.55	60	0.49	60	5.0	200 1/	3,700
M19500/619-22Z	0.74	75	0.70	75	5.0	45	2,600
M19500/619-32Z	0.90	60	0.83	60	1.5	20	1,400
M19500/619-21Z	0.55	10	0.49	10	2.0	15	1,000
M19500/619-31Z	0.75	10	0.63	10	1.0	10	600

1/ T_J = +110°C.

1.5 Type designation. The type designation used under this part shall be formulated as follows:

M19500/619- X Y Z
- Z: Package outline.
- Y: Package size.
- X: Voltage type.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

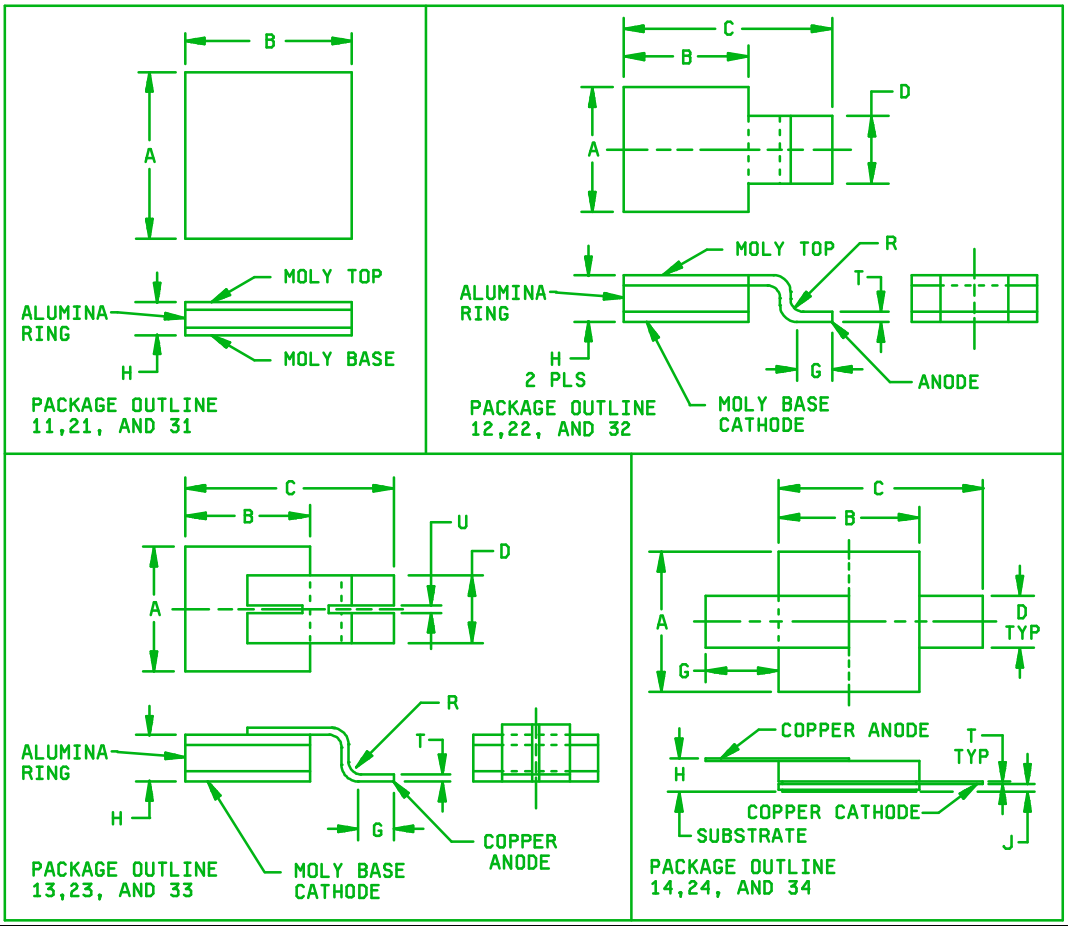
MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Package outline (YZ)	A	B	C	D	G	H	J	R	T	U
11	0.225	0.225	- - -	- - -	- - -	0.075 max	- - -	- - -	- - -	- - -
21	0.315	0.315	- - -	- - -	- - -	0.075 max	- - -	- - -	- - -	- - -
31	0.410	0.410	- - -	- - -	- - -	0.075 max	- - -	- - -	- - -	- - -
12	0.225	0.225	0.375	0.125	0.060	0.075 max	- - -	0.020	0.020	- - -
22	0.315	0.315	0.465	0.170	0.060	0.075 max	- - -	0.020	0.020	- - -
32	0.410	0.410	0.550	0.200	0.060	0.075 max	- - -	0.020	0.020	- - -
13	0.225	0.225	0.375	0.150	0.060	0.090 max	- - -	0.020	0.015	0.015
23	0.315	0.315	0.465	0.150	0.060	0.090 max	- - -	0.020	0.015	0.015
33	0.410	0.410	0.560	0.150	0.060	0.090 max	- - -	0.020	0.015	0.015
14	0.225	0.225	0.405	0.150	0.190	0.138 max	0.037	- - -	0.012	- - -
24	0.315	0.315	0.495	0.150	0.190	0.138 max	0.037	- - -	0.012	- - -
34	0.410	0.410	0.590	0.150	0.190	0.138 max	0.037	- - -	0.012	- - -
Tolerances	±0.010	±0.010	±0.020	±0.010	±0.010	- - -	±0.005	±0.005	±0.002	±0.005

NOTE: All dimensions are in inches.

FIGURE 1. Physical dimensions.

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3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 herein.

3.4.1 Base, lid, and lead finish. Base and lid material shall be Molybdenum. Lead finish shall be tin-plated or solder-dipped. When a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Internal construction. Multiple chip construction shall not be permitted.

3.5 Marking. Devices shall be marked in accordance with MIL-PRF-19500, abbreviation of the part can be as follows JVM619-XYZ.

3.6 Electrostatic discharge protection. The devices covered by this specification shall be classified for ESD sensitivity in accordance with MIL-PRF-19500 and MIL-STD-750, method 1020, and shall be handled, marked, and packaged accordingly.

3.7 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.8 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3 herein.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

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4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see appendix E, table IV of MIL-PRF-19500)	Measurements
	JANTX and JANTXV levels
Thermal response 1/	MIL-STD-750, method 3101 (see 4.5.3)
9	Not applicable
10	MIL-STD-750, method 1038, test condition B T_C = +100°C
11	Subgroup 2 of table I herein
12	Not applicable
13	Subgroup 2 of table I herein; $\Delta I_R \leq$ 5.0 mA or 100 percent of initial value, whichever is greater. $\Delta V_F \leq$ .05 volts or ± 100 percent of initial value, whichever is greater.

1/ Shall be performed anytime before screen 10.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with 4.3, screen 13.

4.4.2.1 Group B inspection, appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1037	V_{RWM} = rated PIV, I_O set to maintain T_C = +100°C, T_A = +25°C ± 3°C; $t_{on} = t_{off}$ = 1 minute minimum for 2,000 cycles. No heat sink or forced air shall be permitted.
B4	2075	See 3.3.2.
B5	4081	See 4.5.2.

NOTE: Subgroups 2, 3, and 6, electrical measurements, see 4.3, screen 13.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500. Electrical measurements (end-points) and delta requirements shall be in accordance with 4.3, screen 13).

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4.4.3.1 Group C inspection, appendix E, table V of MIL-PRF-19500.

Subgroup	Method	Conditions
C3	2016	1,500 G's, 0.5 ms.
C6	1037	V_{RWM} = rated PIV, I_O set to maintain T_C = +100°C, T_A = +25°C ± 3°C; t_{on} = t_{off} = 1 minute minimum for 6,000 cycles. No heat sink or forced air shall be permitted. (Test condition D).

NOTE: Subgroups 2, 3, and 6, electrical measurements, see 4.3, screen 13.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions for subgroup testing in table II herein. Electrical measurements shall be in accordance with 4.3, screen 13.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Thermal resistance. Thermal resistance measurement shall be conducted in accordance with test method 4081 of MIL-STD-750. The maximum limit of $R_{\theta JC}$(max) shall be in accordance with 1.3.

4.5.3 Thermal response. The ΔV measurement shall be performed in accordance with method 3101 of MIL-STD-750. The ΔV conditions (I_H and V_H) and maximum limit shall be derived by each vendor from the thermal response curves. The following parameter measurements shall apply:

a. Measuring current (I_M) ..... +100 mA.

b. Heating current (I_H) ..... 20 A minimum.

c. Heating time (t_H) ..... 100 ms.

d. Measurement time delay (t_{MD}) ..... +10 to 100 µs.

e. Sample window time (t_{SW}) ..... 5 µs maximum.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 2/		Unit
	Method	Conditions 2/		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	t_p = 300 µs I_F = Column 9	V_{FM}		Col 8	V (pk)
Reverse current	4016	V_R = Column 2	I_{R1}		Col 12	mA dc
Subgroup 3						
High temperature operation:		T_C = +125°C				
Reverse current	4016	V_R = Column 2	I_{R2}		Col 13	mA dc
Subgroup 4						
Capacitance (per leg)	4001	V_R = 5 V; 1 kHz ≤ f ≤ 1 MHz	C_J		Col 14	pF
Subgroup 5						
Surge current	4066	I_{FSM} = rated (Column 4), 10 surges of 8.3 ms each at 1 minute intervals; V_{RWM} = rated PIV (Column 2)				
Electrical measurements		See 4.3, screen 13				

1/ For sampling plan, see MIL-PRF-19500.
2/ See 1.3 and 1.4 for conditions and limits.

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TABLE II. Group E inspection (all quality levels) for qualification only. 1/

Inspection	MIL-STD-750		Sample size
	Method	Conditions	
Subgroup 1 Thermal shock Electrical measurement	1051	-55°C to +150°C, 500 cycles See 4.3, screen 13	22 c = 0
Subgroup 2 2/ Steady-state reverse bias Electrical measurement	1038	Test condition A; 1,000 hours; T_J = +150°C; V_R = 80-85 percent rated PIV. See 4.3, screen 13	22 c = 0
Subgroup 3 Not applicable			
Subgroup 4 2/ Thermal resistance, junction to lead	3101 or 4081	$R_{\theta JC}$ = Column 5 (°C/W maximum) See 4.5.2	11 c = 0
Subgroup 5 Not applicable			
Subgroup 6 ESD	1020		11 c = 0

1/ For initial design verification only (one time testing).
2/ A separate sample may be pulled for each test.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, AV, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/619A	**2. DOCUMENT DATE (YYMMDD)** 971111

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, SCHOTTKY RECTIFIERS, JAN, JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial (614)692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000.	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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